Attributes in Domain Modeling

Definition

An **attribute** is a logical data value of a conceptual class, used to capture information needed for current scenarios or requirements.

When to Show Attributes

Guideline

Include attributes suggested or implied by requirements (e.g., use cases). Examples:

- **Sale**: dateTime
- **Store**: name, address
- **Cashier**: ID
- **Receipt**: date, time, store info, cashier ID

UML Notation for Attributes

UML Attribute Notation

- Shown in the second compartment of the class box.
- Full syntax: `visibility name : type multiplicity = default {property-string}`
- Common conventions:
  - Default visibility: private (-)
  - Derived attribute: prefix with '/' (e.g., '/total')
  - Multiplicity: indicates optional presence or collection size (e.g., 'middleName : [0..1]')

Derived Attributes

Derived Attributes

- Attributes calculable from other information.
- Example: **Sale /total** derived from **SalesLineItems**.
- Quantity entered in **SalesLineItem** may be derived from the multiplicity of Item instances.

Guidelines for Attribute Types

Attribute Types

- Prefer **primitive data types**: Boolean, Number, Date, Time, String, Character.
- Specialized types: Address, Color, Geometrics, Phone Number, UPC, SKU, enumerations.
- Do not use complex classes as attribute types; relate with associations instead.
- Example: Cashier uses Register → use association, not attribute ‘currentRegister‘.

Data Type Classes

Data Type Classes

Create a new data type class when:

- It has multiple subparts (e.g., **ItemID**, Address)
- Operations associated (e.g., validation, parsing)
- Has multiple attributes (e.g., Money: amount + currency)
- Represents a quantity with unit (e.g., weight, price)
- It generalizes multiple types (e.g., UPC/EAN → ItemID)

Quantities and Units

Quantities

Numeric quantities should be represented as classes with associated units.

- Example: **Money** class for prices (unit = currency)
- Example: **Weight** class (unit = kg, lb)

Do Not Represent Foreign Keys as Attributes

Guideline

Use **associations** to relate conceptual classes instead of foreign key attributes. Example: Do not use ‘currentRegisterNumber‘ in Cashier; instead, model Cashier–Register association.

Example: POS Domain Attributes

Example

- **Payment**: amount : Money
- **ProductDescription**: description, itemId : ItemID, price
- **Sale**: dateTime
- **SalesLineItem**: quantity
- **Store**: name, address
- **CashPayment**: amountTendered